# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

## ASSESSMENT 2 – Concept Mapping (5 QUESTIONS)

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|---|---|---|---|
| **Course Code:** | ITA05 | **Course Title:** | **Computer Vision** |
| **CO Assessed:** | CO1 | **Assessment:** | Assessment 2: Concept Mapping |
| **Weightage:** | 35% of CIA | **Total Marks:** | (5 Questions × 10 Marks) |
| **CO1 Statement:** | ***Apply the fundamental concepts, image formation models, and image processing techniques to analyze and solve real-world Computer Vision problems.*** | | |
| **Student Name:** | G. Poojitha | **Reg. No.:** | 192425337 |
| **Section / Batch:** | Slot A / 2024 | **Date:** | 5-10-2026 |

**Code of Conduct**

*I* G. Poojitha (192425337) *(NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: G. Poojitha

**QUESTIONS: 5**

**Total Marks: 50**

Annexure A

## Concept Mapping

Q1. An automobile company is developing an autonomous vehicle that uses cameras to detect vehicles, pedestrians, and traffic signs. Objects closer to the camera appear larger, while distant objects appear smaller.
**Prepare a concept map applying the image formation and camera projection model to show how a three-dimensional road scene is converted into a two-dimensional image. Include the relationships between focal length, object distance, image distance, and projected image size.**

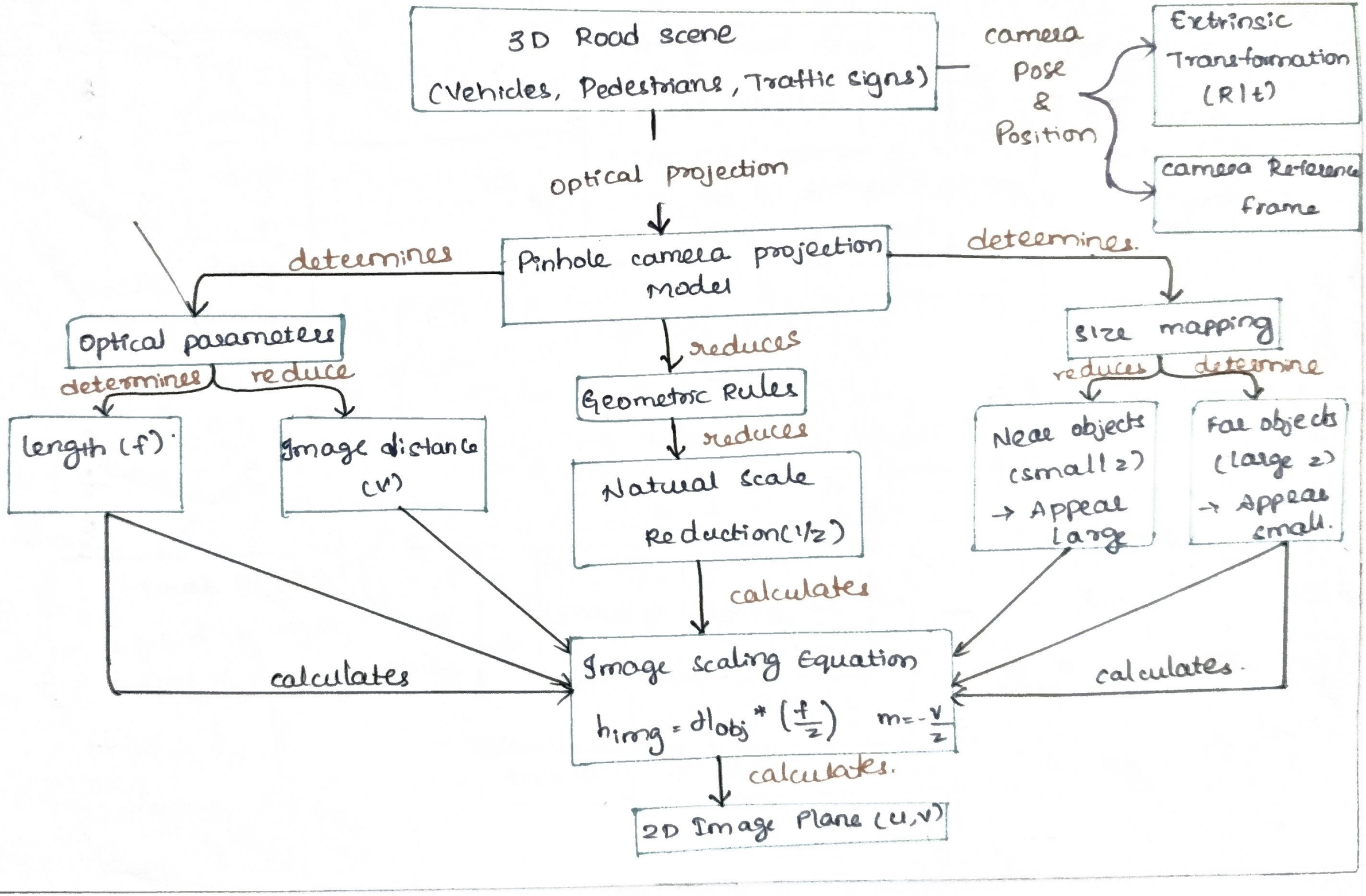
Q2. A manufacturing company uses cameras to measure the dimensions of bottles moving on a conveyor belt. Engineers observe that bottles at different distances from the camera appear to have different sizes.
**Prepare a concept map applying perspective projection to show how focal length, camera position, viewing angle, object distance, and image size are related. Use the concept map to represent how these factors influence the accuracy of bottle measurement.**

Q3. A construction company uses a drone camera to capture images of buildings from different heights and viewing angles. The appearance and measured dimensions of the building change when the drone position is changed.
**Prepare a concept map applying image formation and projection concepts to show the relationship between camera orientation, focal length, perspective projection, camera position, calibration, and the accuracy of building measurements.**

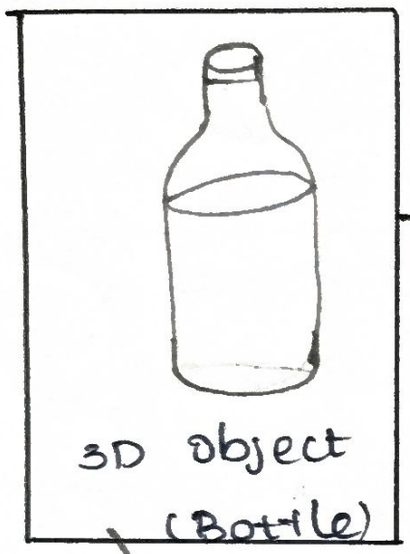
Q4. An industrial company wants to select a suitable projection model for road monitoring and inspection of objects placed on a flat conveyor belt. The engineers are considering perspective and orthographic projection for these applications.
**Prepare a concept map comparing perspective and orthographic projection based on depth, object size, distance, parallel lines, and image formation. Use the concept map to identify the appropriate projection model for each application.**

Q5. A robotics company wants a robot to identify the position and shape of objects placed on a table using a camera. The camera captures only a two-dimensional representation of the three-dimensional objects.
**Prepare a concept map applying the image formation model to show the relationship between a 3D object, camera parameters, projection geometry, focal length, camera orientation, and the resulting 2D image. Show how these concepts help determine the object's position and appearance.**

# 1) Autonomous Vehicle Road Scene Projection.



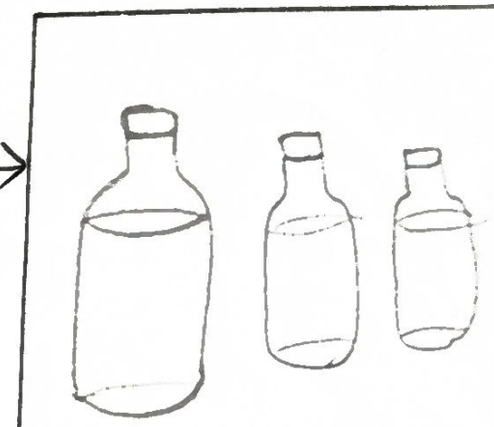
## 2) Manufacturing - Perspective Projection.




Seen by camera

**Camera Parameters**

- Focal length(f)
- camera position (height & distance)
- viewing angle (FOV)




**Perspective projection**

$$h' = \frac{f}{z} \times H$$

($h'$ = image size, $H$ = real size, $z$ = object distance)

**Factors & their Influence**

**Focal length(f)**
- larger f → larger image
- smaller f → smaller image

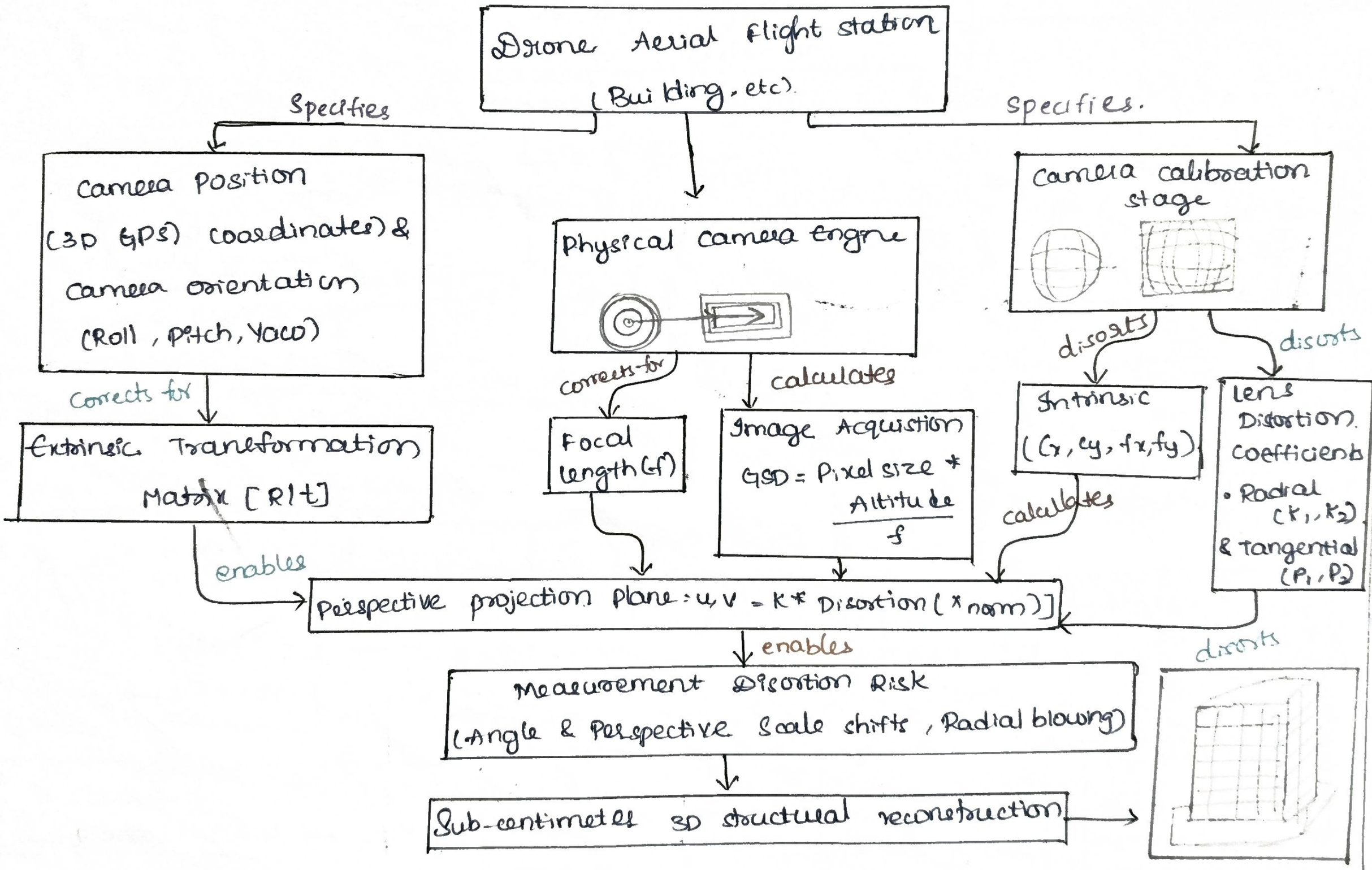
**camera position**
- closer → large image
- Farther → smaller

**Viewing angle**
- wider → more area, smaller object
- Narrow → large object

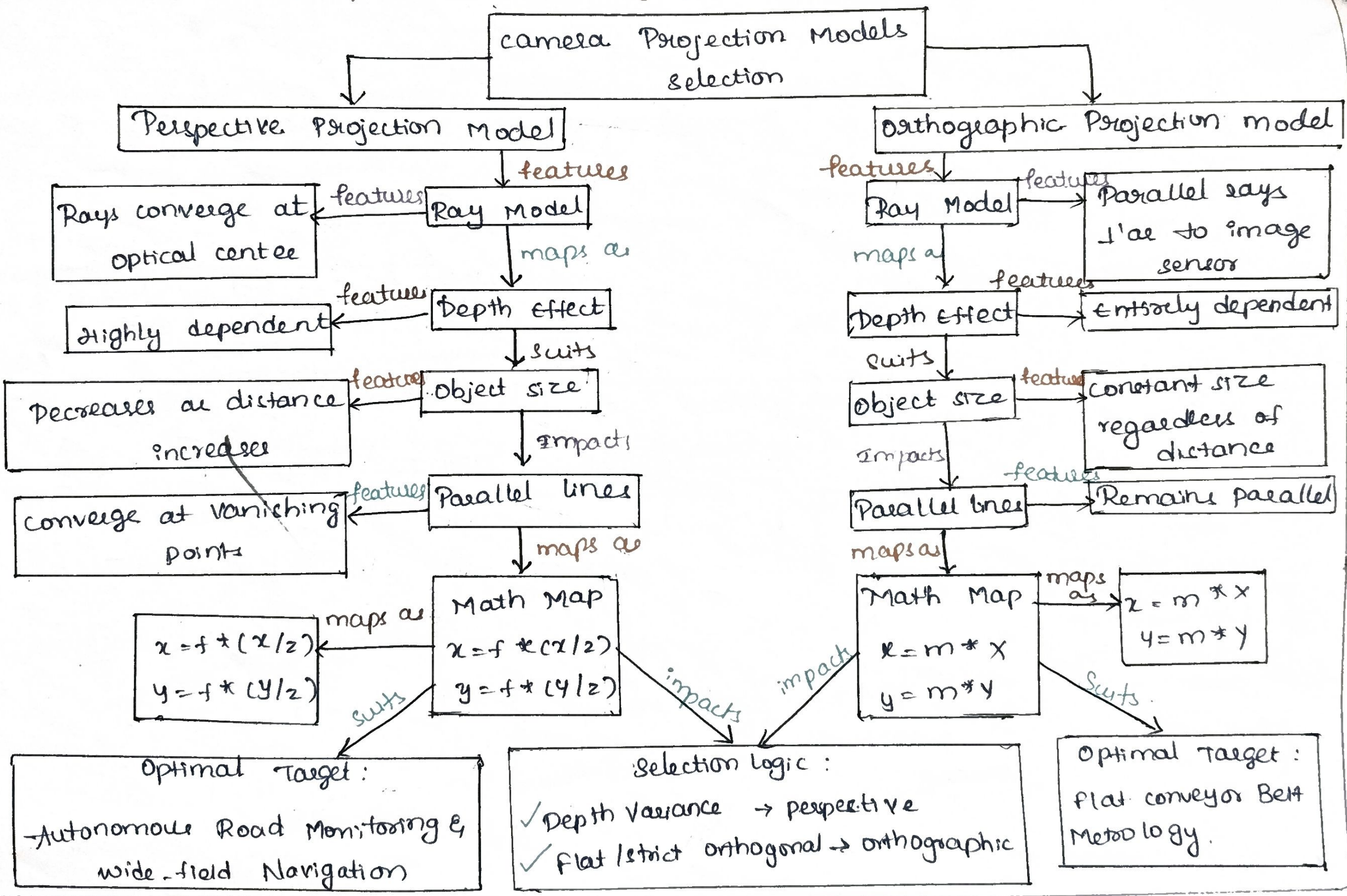
**object distance**
- closer → larger
- Farther → smaller

**Accuracy of Bottle Measurement**

# 3) Drone Camera - Image Formation & Projection



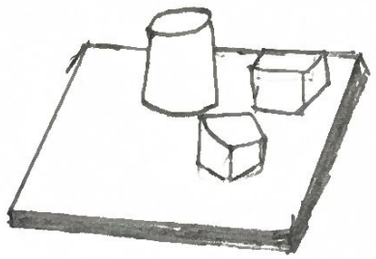
# 4) Perspective Vs. Orthographic Projection select Evaluation



# 5) Robot Vision 3D to 2D shape & Position

3D workpiece on Table $(X_w, Y_w, Z_w)$

transforms → Spatial Extrinsic matrix $[R|t]$



transforms → Pinhole Projection Geometry & Intrinsics



optical mapping

$$z_c \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K \cdot [R|t] \begin{bmatrix} x_w \\ y_w \\ z_w \\ 1 \end{bmatrix}$$

Rotation (Roll, Pitch, Yaw)

Translation $(T_x, T_y, T_z)$

maps coordinates

camera frame $(X_c, Y_c, Z_c)$

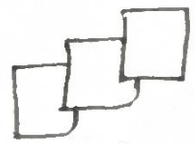
defines optical mapping

Resulting 2D Image plane Representation $(u, v)$

Projects

Foreshortening / Scale Distortion

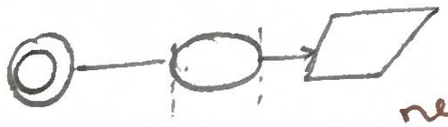
Occlusion Boundaries



-foreground blocking

circular rims

Project ellipse



needs reconstruction

solves for 3D.

Determining 3D position & 6-DOF Pose Estimation

estimate

Stereo camera / Depth Triangulation

Planar constraint

Rubrics for Calculation

| Criteria | Weightage | Level 4 (Exemplary) | Level 3 (Proficient) | Level 2 (Developing) | Level 1 (Beginning) |
|---|---|---|---|---|---|
| **Technical Content Accuracy** | 30 | Highly accurate, indepth, and insightful technical explanation | Technically correct content with adequate depth and clarity | Basic concepts presented with limited accuracy and depth | Content contains major technical errors; concepts poorly understood |
| **Problem Identification** | 25 | Problem rigorously analyzed using appropriate and advanced methods | Problem clearly defined with a logical engineering approach | Problem identified but analysis or methodology is weak | Problem statement unclear or incorrect; no logical approach |
| **Use of Tools** | 20 | Advanced tools, well-analyzed data, and highly effective visuals | Appropriate tools and data used effectively with clear visuals | Limited use of tools and basic data interpretation | Minimal or incorrect use of tools and data; poor visuals |
| **Communication** | 15 | Highly engaging, confident, and audience-focused delivery | Clear, organized, and professional communication | Understandable but lacks clarity, structure, or confidence | Poor organization, unclear speech, ineffective slides |
| **Professionalism** | 10 | Exemplary professionalism, leadership, and ethical awareness | Professional conduct with effective team collaboration | Basic professionalism; uneven team participation | Lacks professionalism; minimal contribution or ethical awareness |

| Faculty In-charge | Course Coordinator | Program Director |
|---|---|---|
| Signature: _____ | Signature: _____ | Signature: _____ |
| Date: 5/10/26 | Date: _____ | Date: _____ |